James Ruse Agricultural High School

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2024

HIGHER SCHOOL CERTIFICATE TRIAL EXAMINATION

Physics

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black pen
- Draw diagrams using pencil
- NESA approved calculators and equipment may be used
- A formula sheet and data sheet and Periodic Table are provided.
- For questions in Section II, **show all relevant working** in questions involving calculations.

Total marks 100

Section I -20 marks (pages 2-13)
- Attempt Questions 1– 20
- Allow about 35 minutes for this part

Section II -80 marks (pages 16-36)
- Attempt Questions 21–35
- Allow about 2 hours and 25 minutes for this part

This paper MUST NOT be removed from the examination room.

Section I

20 marks
Attempt Questions 1-20
Allow about 35 minutes for this section.
Mark your answers on the ANSWER grid in the Answer booklet on page 15.

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

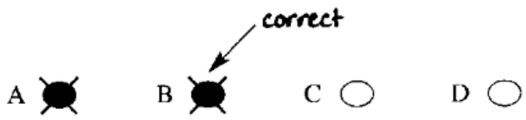
Sample: $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9

A ○ B ● C ○ D ○

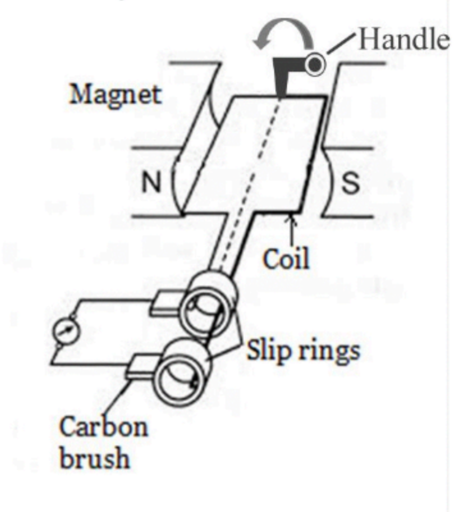
If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ● B ~~●~~ C ○ D ○

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word *correct* and drawing an arrow as follows.



1. What is the name of the following device?



A. DC motor

B. DC generator

C. AC generator

D. AC induction motor

2. An object is moving in a horizontal circle on a string with a constant angular velocity ω. The tension in the string, which is approximately horizontal, is F. If ω is doubled while all the other quantities are kept constant, what will the tension in the string become?

A. 4F

B. 2F

C. F/2

D. F/4

3. If an astronaut orbiting the Earth near its surface were to stand on a set of bathroom scales, why would there be a reading of zero?

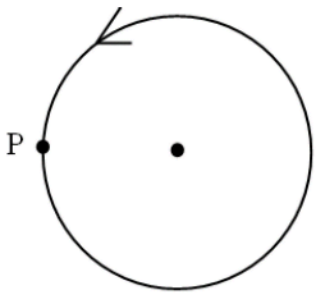
A. The astronaut's mass becomes zero.

B. The strength of the gravitational field is zero.

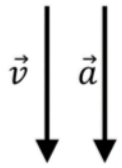
C. The pull of the Earth is balanced by the pull of the Moon.

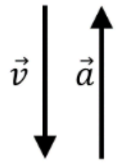
D. The astronaut falls towards the Earth at the same rate as the space capsule.

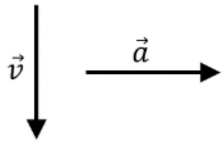
4. An object is executing uniform circular motion, as shown in the figure below.

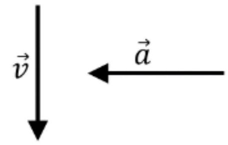


At position P, which of the following shows the directions of instantaneous velocity and acceleration?

A. 

B. 

C. 

D. 

5. Iodine–131 is a radioisotope used in nuclear medicine. It has a half-life of 8 hours and eventually becomes a stable isotope Xenon-131 according to the following equation:

$$^{131}_{53}\mathrm{I} \rightarrow {}^{131}_{54}\mathrm{Xe} + \gamma$$

24.0 mg of Iodine–131 is injected in the body of a patient.

How many milligrams of Xenon-131, produced as a result of this reaction, is present in the body after 16 hours?

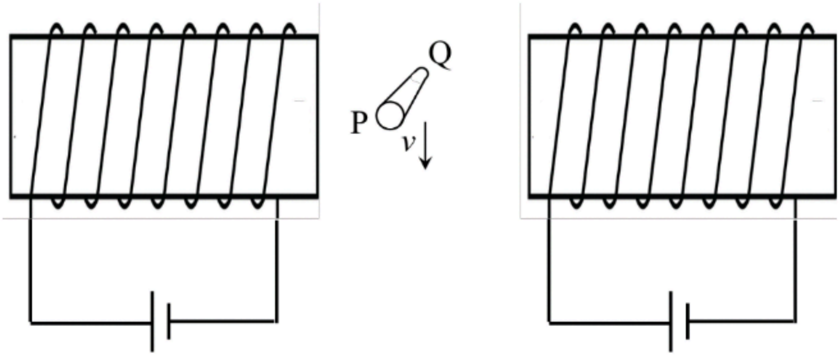
A. 6.0

B. 12.0

C. 18.0

D. 24.0

6. A horizontal metal cylinder PQ, perpendicular to the page, moves down between two solenoids, as shown below:



Which of the following correctly show the direction of the magnetic field between the solenoids and the charge at P?

	Direction of magnetic field	*Charge at P*
A.	left	positive
B.	left	negative
C.	right	positive
D.	right	negative

7. A radioactive isotope decays by alpha and beta particle emission. Over a number of decays, the nucleon number decreases by 4 while the proton number is unchanged.

How many alpha and beta particles have been emitted?

A. 2 alpha particles and 1 beta particle

B. 1 alpha particle and 2 beta particles

C. 1 alpha particle and 1 beta particle

D. 2 alpha particles and 2 beta particles

8. Two students are discussing the advantages of magnetic brakes.

Student A says that because the magnetic force is proportional to velocity they provide a smoother braking action.

Student B says that because magnetic fields do no work, the brakes produce no heat.

Which of the students, if any, is/are correct?

A. Student A only

B. Student B only

C. Both student A and student B

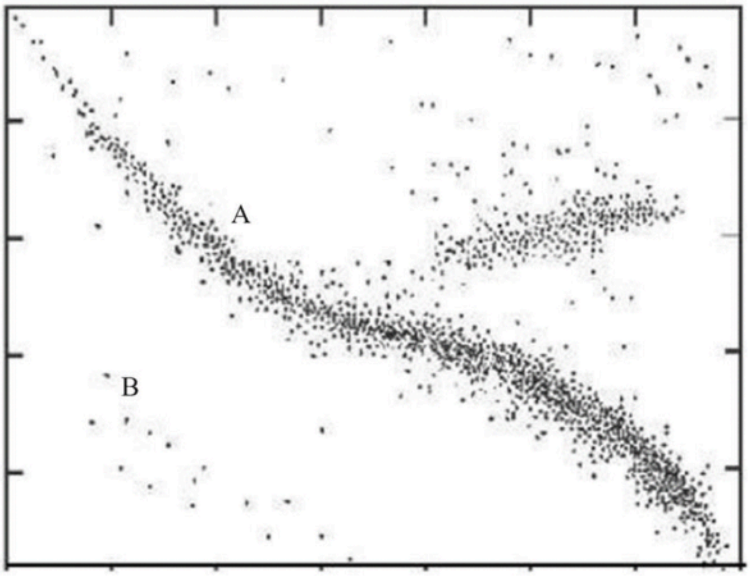
D. Neither student A nor student B

9. Control rods and moderators play a key role in controlling the rate of fission in nuclear reactors.

Which row of the table correctly shows how this is achieved?

	Control Rod	*Moderator*
A.	By absorbing neutrons.	By slowing down fast neutrons to thermal speeds.
B.	By absorbing neutrons.	By speeding up slow neutrons so that a chain reaction is sustainable
C.	By increasing the capture cross-section of neutrons.	By speeding up slow neutrons so that a chain reaction is sustainable
D.	By increasing the capture cross-section of neutrons.	By slowing down fast neutrons to thermal speeds.

10. This is a Hertzsprung-Russell (H-R) diagram showing many star plots. The axes are not labelled.



Which of the following statements is true of the luminosity and temperature of stars A and B?

A. The luminosity of star A is lower while the temperature of star B is higher.

B. The luminosity of star A is lower while the temperature of star B is lower.

C. The luminosity of star A is higher while the temperature of star B is higher.

D. The luminosity of star A is higher while the temperature of star B is lower.

11. At what angle, θ should two polaroid sheets be placed to reduce the intensity by ninety percent?

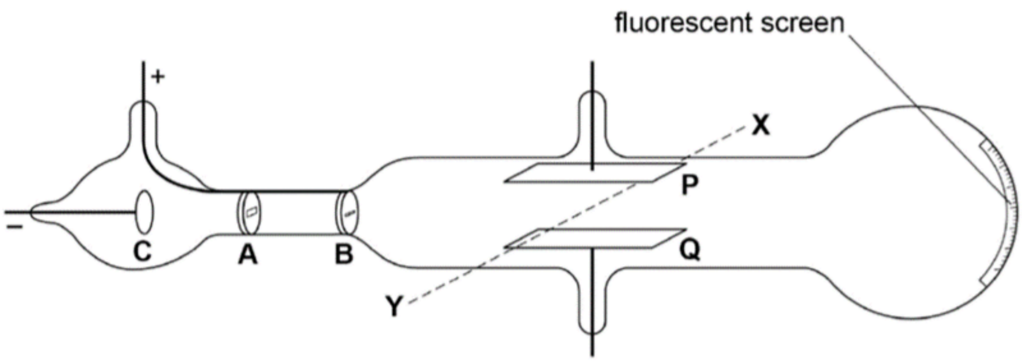
A. $\theta = \cos^{-1}\sqrt{0.8}$

B. $\theta = \cos^{-1}\sqrt{0.2}$

C. $\theta = \cos^{-1}\sqrt{0.9}$

D. $\theta = \cos^{-1}\sqrt{0.1}$

12. The diagram shows a discharge tube used by Thomson to investigate cathode rays. When the connections are made to a high voltage power supply, cathode rays are produced at C and accelerated across AB and through parallel plates P and Q, connected to a second power supply, with P positively charged. The rays then hit the centre of the fluorescent screen, when a magnetic field is positioned (in the XY direction) between the plates P and Q.



Which row of the table shows the aim of this experiment and the correct direction of the applied magnetic field?

	Aim of the experiment	*Direction of magnetic field*
A.	To determine the charge of the electron	From X to Y
B.	To determine the charge of the electron	From Y to X
C.	To determine the charge to mass ratio of the electron	From X to Y
D.	To determine the charge to mass ratio of the electron	From Y to X

13. Two satellites orbit a planet. One of the satellites has an orbital radius of 2.3×10^{6} m and a period of 7.6×10^{4} s. The other satellite has an orbital period of 1.3×10^{5} s. What is the orbital radius of this second satellite?

A. 1.3×10^{6} m

B. 3.3×10^{6} m

C. 3.9×10^{6} m

D. 5.1×10^{6} m

14. A 500 kg satellite is initially in a circular orbit with a radius of 10 000 km around the Earth. The satellite is then moved to a new circular orbit with a radius of 20 000 km.

What is the change in the gravitational potential energy of the satellite?

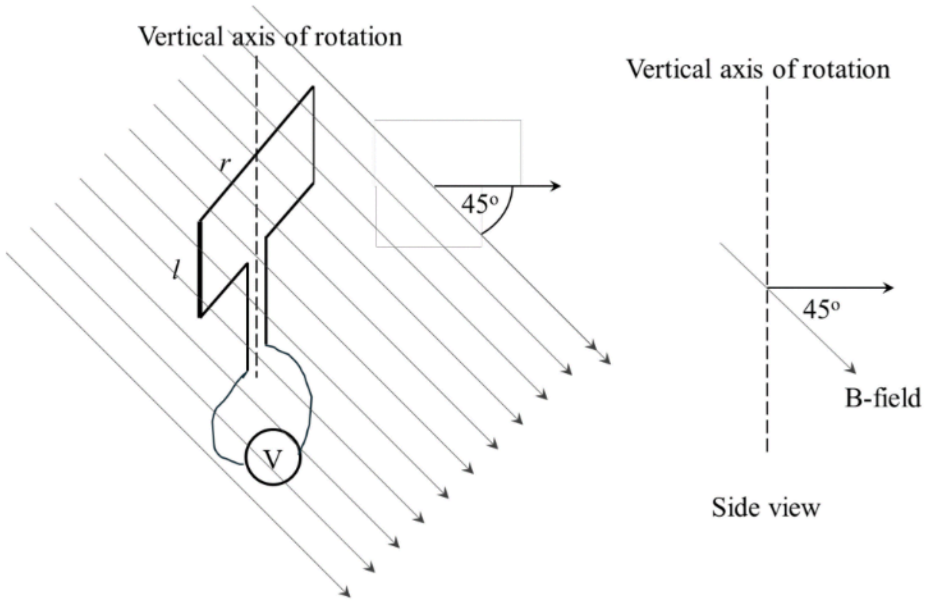
A. 1.5×10^{3} J

B. 1.5×10^{9} J

C. 1.0×10^{10} J

D. 1.0×10^{13} J

15. A coil of wire is rotating about a vertical axis in a magnetic field that is directed 45° below the horizontal:



Which set of changes will result in the largest reading on the voltmeter?

A. Double *r* and halve *l*.

B. Increase both *l* and *r* by 50 %.

C. Rotate the magnetic field to vertical and double *r*.

D. Rotate the magnetic field to horizontal and double *l*.

16. Two stars, A and B, are equally bright as seen from the Earth. Stars A and B are 10 and 15 parsecs away from Earth respectively. One parsec = 3.26 light years.

Which of the following statements about the luminosities of stars A and B (L_A and L_B) are correct?

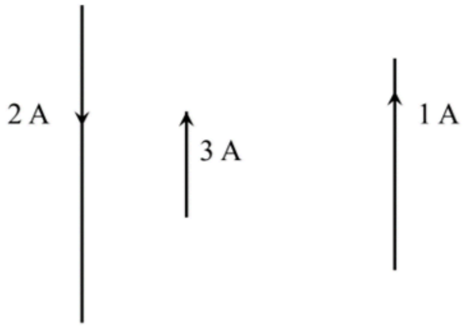
A. $L_A = \frac{3}{2} L_B$

B. $L_A = \frac{2}{3} L_B$

C. $L_A = \frac{4}{9} L_B$

D. $L_A = \frac{9}{4} L_B$

17. Three wires carry currents, as shown in the following scale diagram.



How many of the wires experience a net force to the left?

A. 0

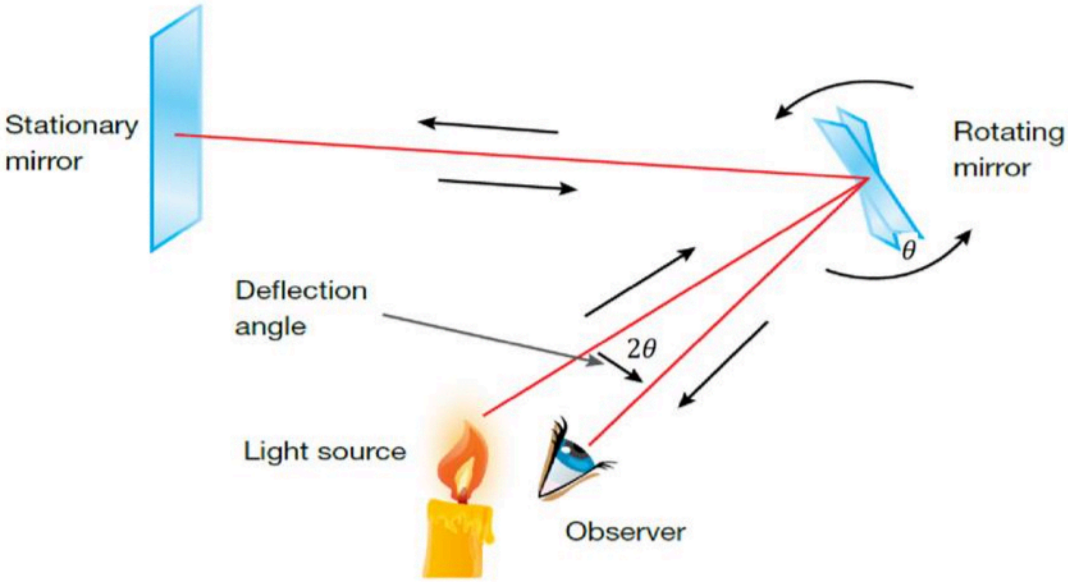
B. 1

C. 2

D. 3

18. Foucault measured the speed of light by focusing light onto a rotating mirror, which reflected it onto a stationary mirror, which in turn reflected it back to a fixed observer.

Whilst the light travelled to the fixed mirror and back, the rotating mirror rotated through an angle, θ. By measuring the angle, θ and with a known angular velocity, ω of the mirror, Foucault was able to calculate a value for the speed of light.



What would be the easiest adjustment to make to the experimental design to reduce the uncertainty in the measured value of the speed of light?

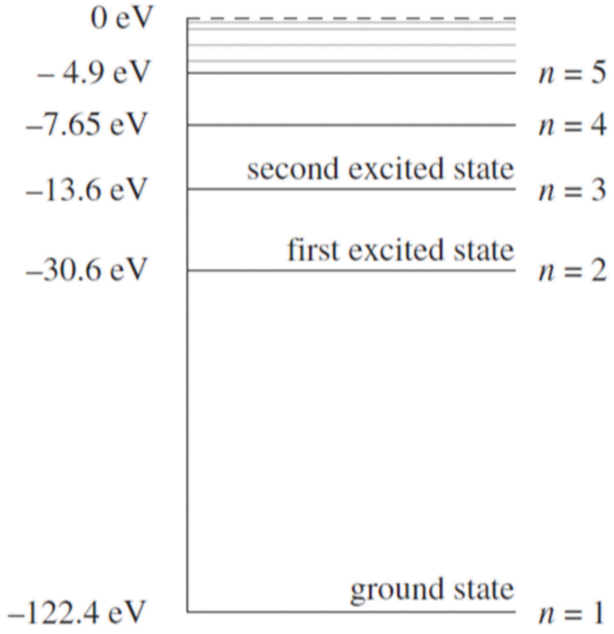
A. Increase the distance between the mirrors.

B. Decrease the distance between the mirrors.

C. Increase the rotational velocity of the rotating mirror.

D. Decrease the rotational velocity of the rotating mirror.

19. The energy level diagram for a particular element is shown below.



Which of the following is inconsistent with the wavelength of a photon emitted from an atom of the element during an electron transition?

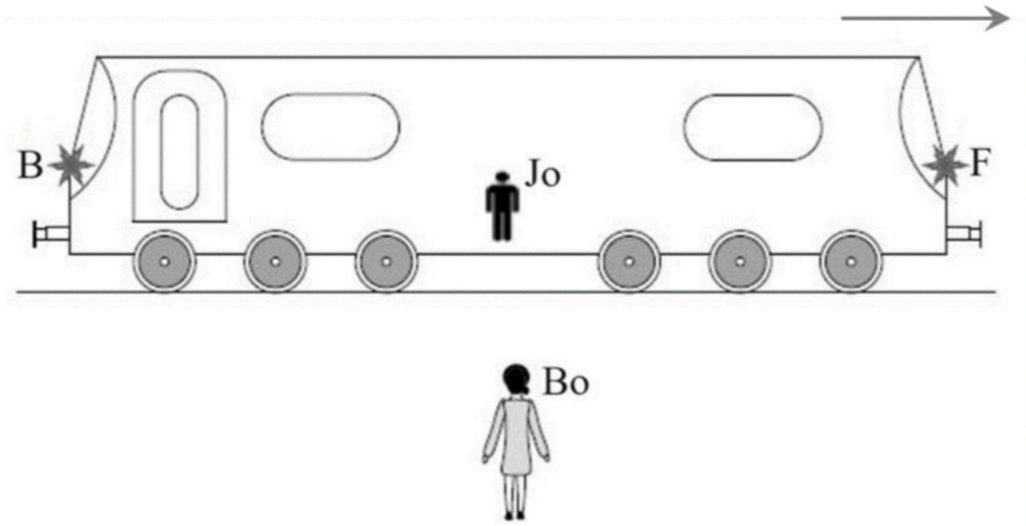
A. 48.3 nm

B. 66.4 nm

C. 72.6 nm

D. 169.0 nm

20. Jo is in the centre of a train carriage travelling close to the speed of light to the right. Lamps F and B located at the front and back of the train respectively are switched on simultaneously as Jo passes an observer Bo who is on the platform equidistant from the lamps when the light is emitted.



According to Bo, which of the following statements is correct in terms of the perceived arrival times of the flashes of light?

A. The lights turn on simultaneously for Jo.

B. The lights turn on simultaneously for Bo.

C. The light at the back turns on before the light at the front for Jo.

D. The light at the front turns on before the light at the back for Bo.

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James Ruse Agricultural High School

Please write your student number NEATLY in the boxes.

Mark

JRAHS Physics Trial 2024

Section I **Multiple Choice Answers**

1.	A ○	B ○	C ○	D ○
2.	A ○	B ○	C ○	D ○
3.	A ○	B ○	C ○	D ○
4.	A ○	B ○	C ○	D ○
5.	A ○	B ○	C ○	D ○
6.	A ○	B ○	C ○	D ○
7.	A ○	B ○	C ○	D ○
8.	A ○	B ○	C ○	D ○
9.	A ○	B ○	C ○	D ○
10.	A ○	B ○	C ○	D ○
11.	A ○	B ○	C ○	D ○
12.	A ○	B ○	C ○	D ○
13	A ○	B ○	C ○	D ○
14.	A ○	B ○	C ○	D ○
15.	A ○	B ○	C ○	D ○
16.	A ○	B ○	C ○	D ○
17.	A ○	B ○	C ○	D ○
18.	A ○	B ○	C ○	D ○
19.	A ○	B ○	C ○	D ○
20.	A ○	B ○	C ○	D ○

Section II

80 marks
Attempt Questions 21-35
Allow about 2 hours and 25 minutes for this section.

Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.

Show all relevant working in questions involving calculations.

Extra writing space is provided at the back of this booklet. If you use this space, clearly indicate which question you are answering.

Question 21 (4 marks)

Artificial satellites monitor weather conditions on Earth and are used for surveillance and communications. Such satellites may be placed in a geostationary orbit.

(a) Identify TWO features of a geostationary orbit. **2**

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(b) Explain what happens to the speed of a satellite when it moves to a lower orbit. **2**

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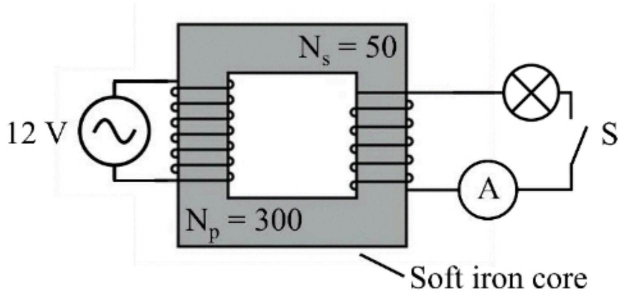
Question 22 (5 marks)

Consider the model of an ideal transformer shown below:



(a) When the switch S is closed, the ammeter reads 0.23 A. **3**

Calculate the power dissipated in the light bulb.

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(b) Explain ONE way in which the model could be improved to more closely approximate an ideal transformer. **2**

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Question 23 (7 marks)

A stationary nucleus of uranium-232 decays into thorium-228 by means of alpha decay. The nuclear masses of each are given below:

Isotope	*Mass (u)*
uranium-232	232.0317
thorium-228	228.0287
helium-4	4.0026

(a) Calculate the energy released in this decay in MeV. **2**

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(b) Calculate the ratio of kinetic energy of the alpha particle to the kinetic energy of the thorium nucleus. **3**

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(c) Explain why the uranium-232 nucleus decays by emitting an alpha particle instead of 2 individual neutrons and 2 individual protons. **2**

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Question 24 (3 marks)

Explain why the period of a satellite in orbit around the Earth cannot be less than 85 minutes. **3**
Include a calculation to support your response.

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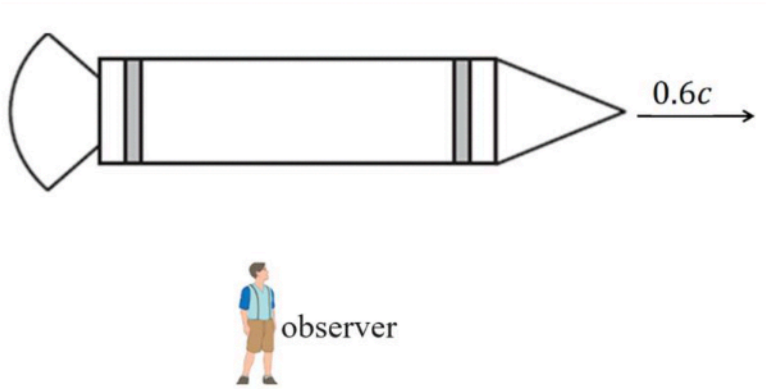
Question 25 (2 marks)

A spacecraft travels at a uniform speed of $0.6c$ within the Earth's gravitational field.
An observer on the ground measures the length of the spacecraft to be 80 m.



Calculate the length of the spacecraft as measured by the astronaut on board. **2**

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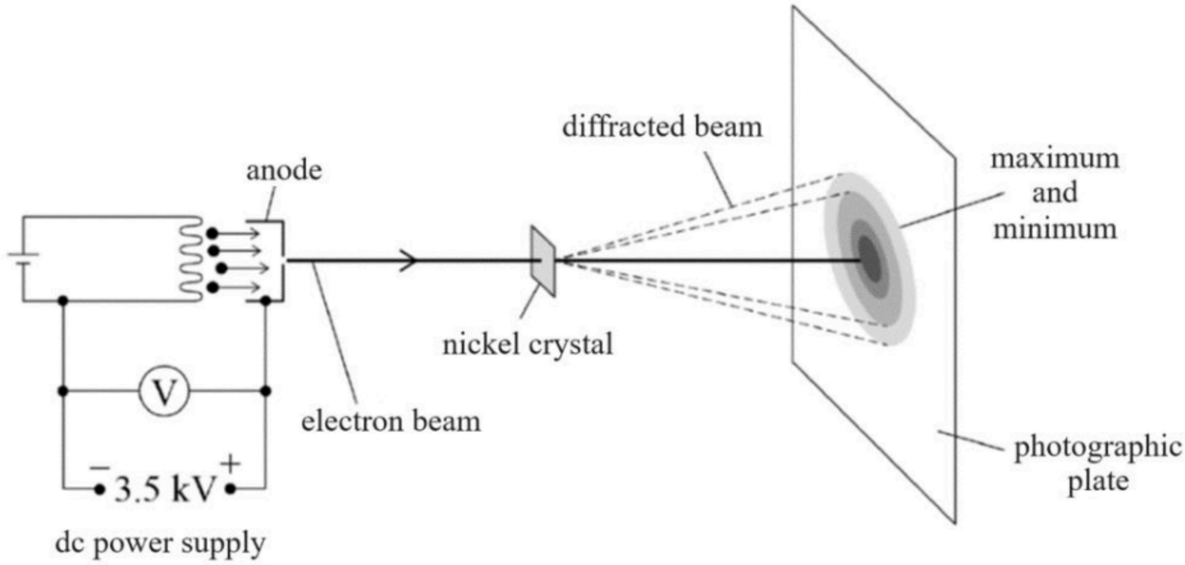
Question 26 (6 marks)

The figure shows part of the apparatus used to investigate de Broglie's matter waves.

Electrons were accelerated through a potential difference to form a beam which was then incident on a thin nickel crystal.

Regions of maximum and minimum intensity were formed on a photographic film behind the crystal.



(a) Show that the de Broglie wavelength of the electrons are given by **3**

$$\lambda = \frac{h}{\sqrt{2mqV}}$$

where V is the voltage of the power supply, m and q the mass and charge of the electron respectively.

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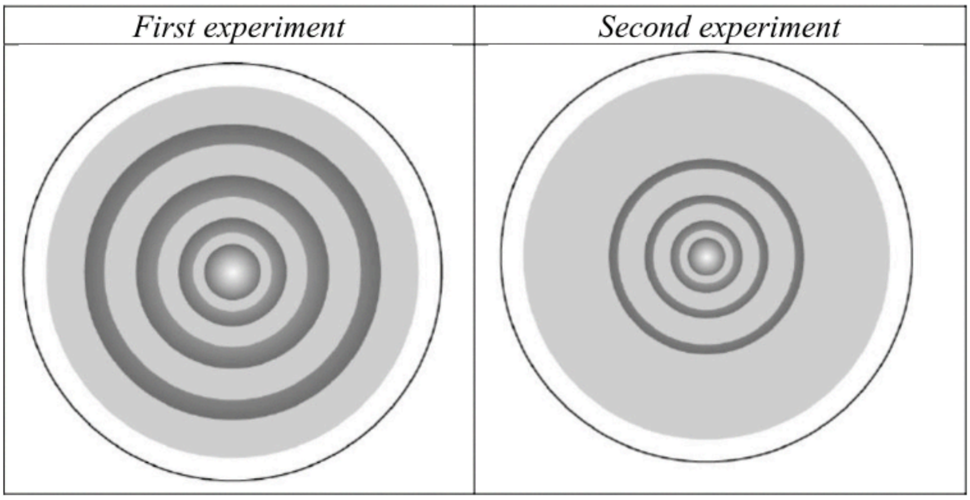
Question 26 continues on page 21

Question 26 (continued)

(b) Determine whether the voltmeter reading of 3.5 kV is consistent with a de Broglie wavelength of about 0.02 nm for the electrons in the beam. **1**

(c) The experiment is repeated using a similar arrangement to that shown above. The diagram below compares the result of the two experiments. **2**

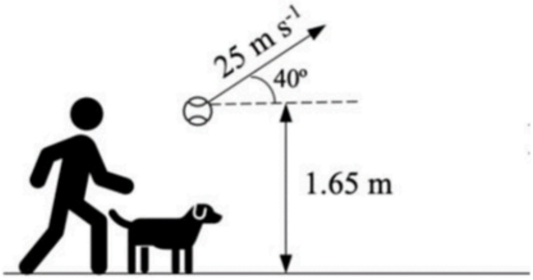


State and explain one independent change that could be made to the arrangement of the first experiment to produce the results shown for the second experiment.

End of Question 26

Question 27 (6 marks)

A person launches a ball upward at a 40º angle with an initial velocity of 25 ms^{-1} from a height of 1.65 meters above the ground.



(a) Calculate the time it takes for the ball to return to the initial height from which it was launched. **2**

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(b) Simultaneously, a dog, initially at rest beside the person, starts running in the same direction as the ball's path with a constant acceleration. **4**

Determine the dog's acceleration if the dog pounces on the ball precisely when it lands on the ground.

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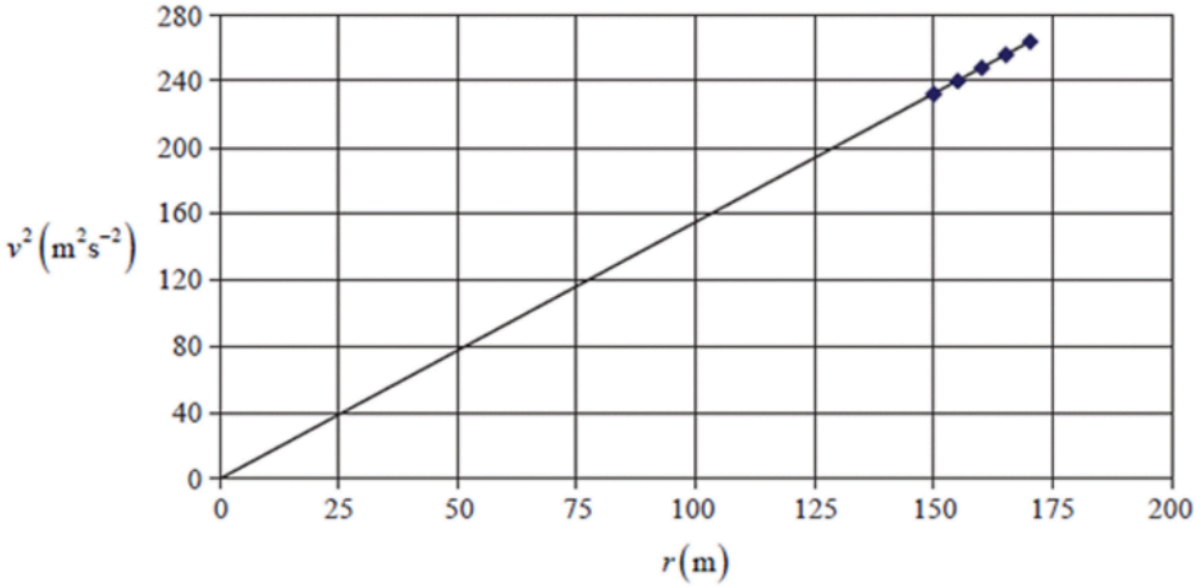
Question 28 (3 marks)

A motor-racing track has banked quarter-circle turns. Racing cars travel around the turns at different radii and speeds. Data are collected when the centripetal acceleration is not caused by the frictional force on the tyres. The graph below shows the data collected.



Using the graph, calculate the banking angle of the turns. **3**

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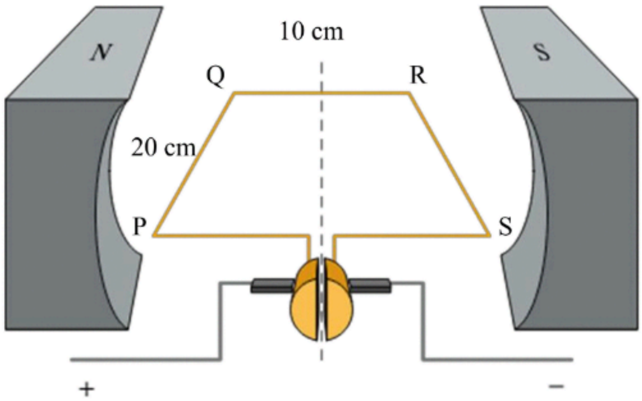
Question 29 (8 marks)

A student was investigating a DC motor with 50 turns of wire in the 20 cm × 10 cm horizontal rectangular coil PQRS by applying a vertical force to one of the sides of the coil to stop the motor from turning.



(a) Annotate the diagram to indicate where this force could be applied. **1**

The student varied the current and measured the force required to keep the coil stationary, as shown below:

Current (A)	Force (N)
0.5	0.011
1.0	0.026
2.0	0.047
3.0	0.068
4.0	0.097

(b) Graph the data on the axes on page 25. **4**

Question 29 continues on page 25

(c) Use the graph to calculate the strength of the magnetic field. **3**

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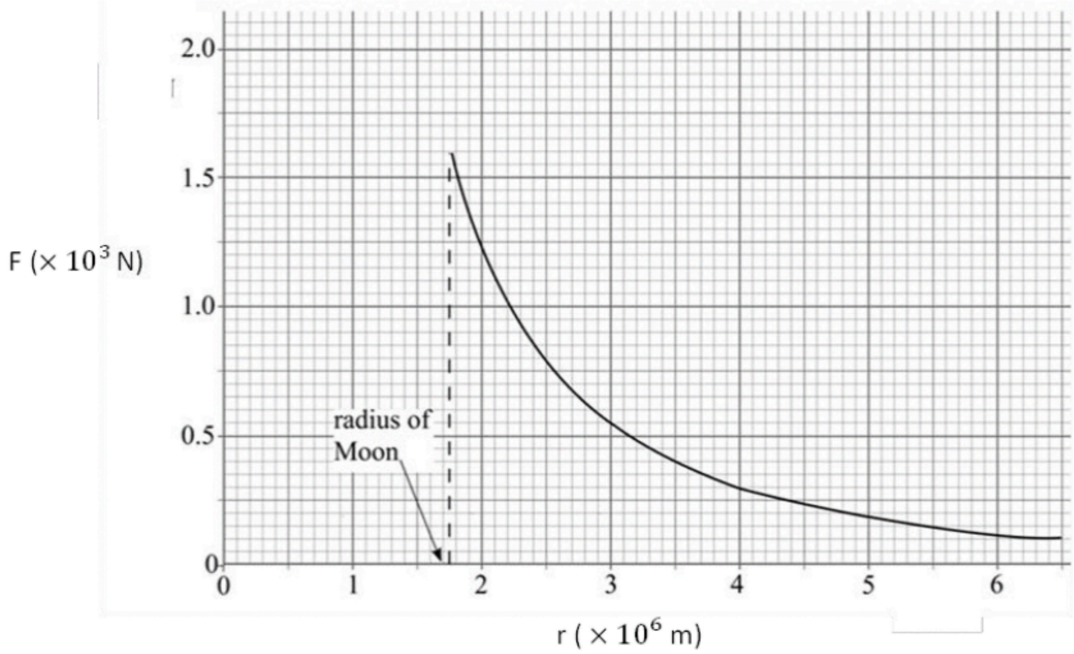
End of Question 29

Question 30 (4 marks)

The graph shows how the gravitational force F between a 1000 kg mass and the Moon varies with the distance r from the Moon's centre for points above its surface.



(a) Use data from the graph to determine the mass of the Moon. **2**

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(b) Calculate the speed at which the 1000 kg mass should be thrown vertically upwards from the surface of the Moon if it is to escape from the Moon's gravitational field. **2**

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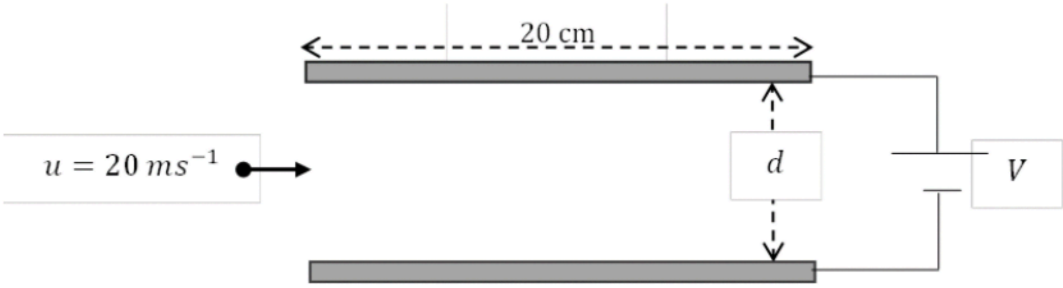
Question 31 (5 marks)

An inkjet printer operates by accelerating charged droplets of ink onto the printer paper. The ink drops are deflected by two plates across which a voltage is applied.



An ink drop of mass $m = 2.0 \times 10^{-10}$ kg and charge $q = 1.0 \times 10^{-13}$ C passes through two plates with an initial horizontal velocity, $u = 20$ m s^{-1}. The plates, a distance $d = 5$ mm apart, are connected to a DC voltage source, $V = 10\ 000$ V.

Ignore the effects of gravity.

(a) Show that the vertical acceleration of the ink drop is given by $a = \frac{Vq}{md}$. **2**

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(b) Find the vertical deflection of the ink drop. **3**

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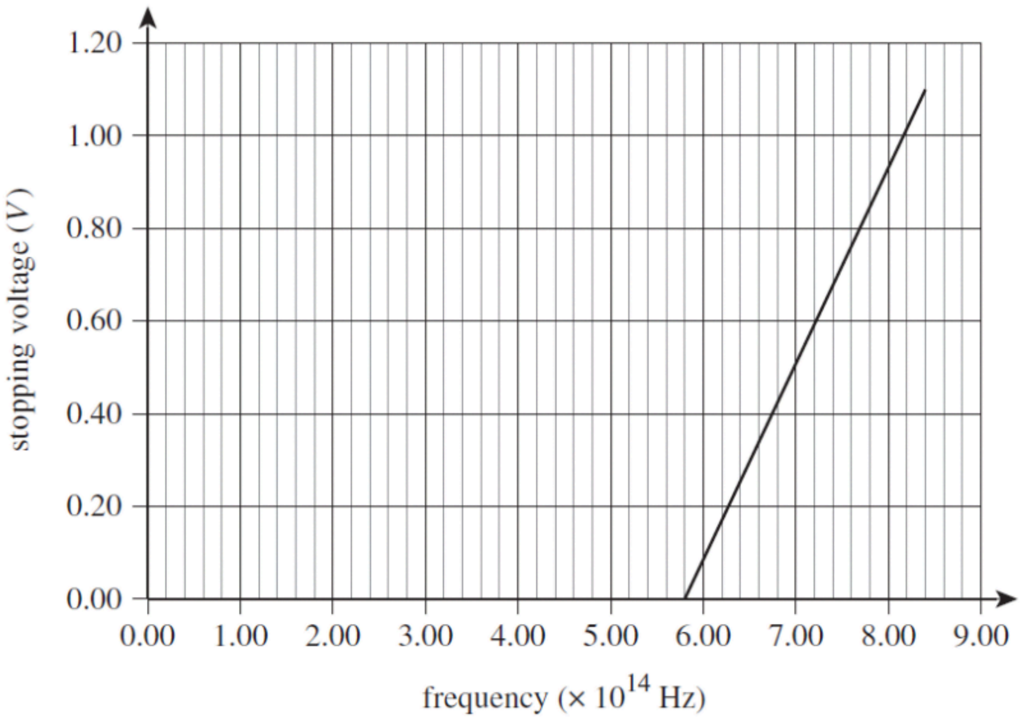
Question 32 (6 marks)

In an experiment involving the photoelectric effect, the results for the stopping voltage versus the light frequency for a particular metal is shown below.



(a) Determine the stopping voltage needed if light of frequency 1.20×10^{15} Hz is used. **3**

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Question 32 continues on page 29

Question 32 (continued)

(b) On the axes provided on page 28, sketch the graph of the results if a metal whose work function of 1.95 eV is used instead. **3**

Any relevant calculations used to assist in sketching should be shown below.

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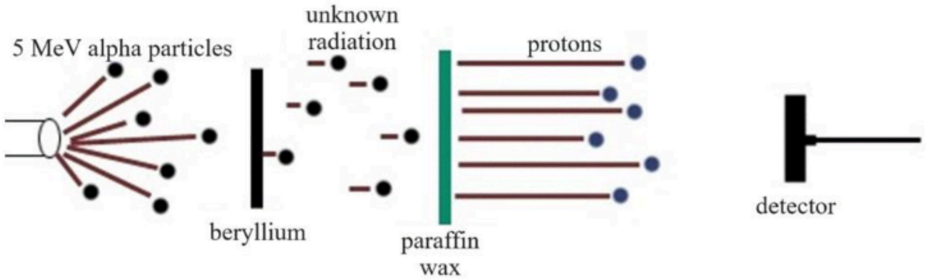
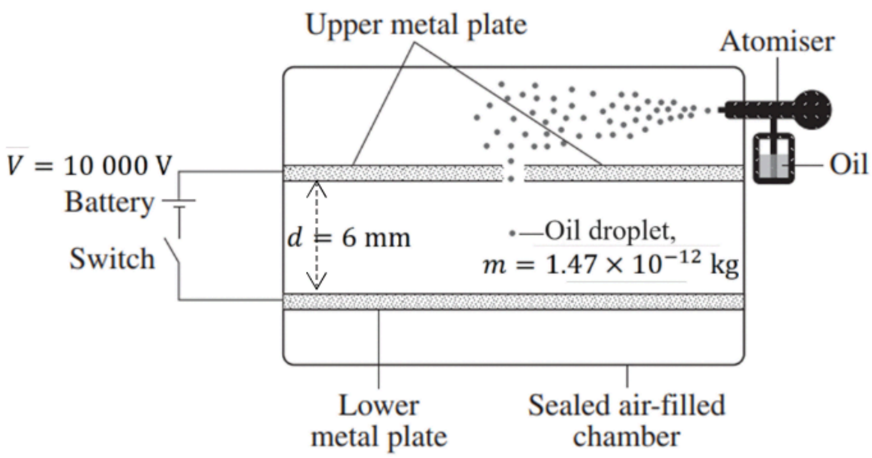
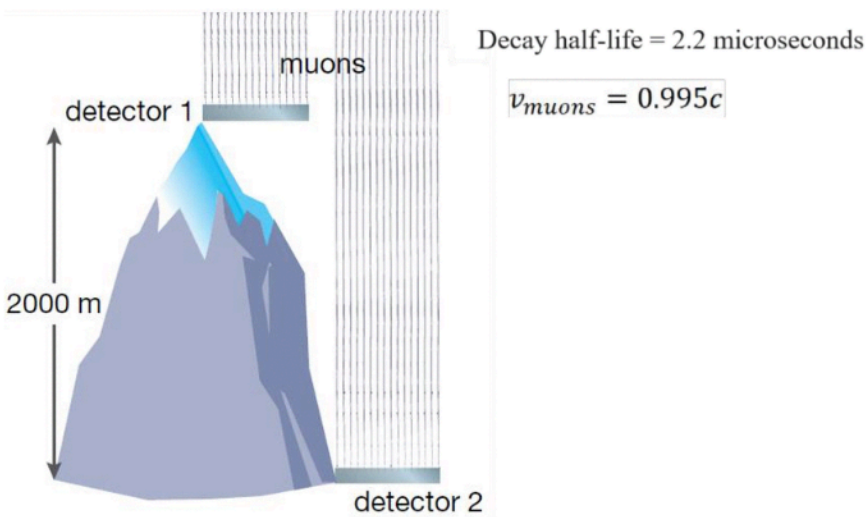
End of Question 32

Question 33 (9 marks)

Experiments conducted by Millikan and Chadwick as well as cosmic-origin muons played a pivotal role in our understanding of the structure of matter and the laws of physics.

Experimental setup	*Data*
	Energy of alpha particles = 5 MeV
	$V = 10\,000$ V $m = 1.47 \times 10^{-12}$ kg Distance between plates, $d = 6$ mm Oil droplet balanced between the plates
	$t_{\frac{1}{2}} = 2.2\ \mu s$ $v_{\mu} = 0.995c$

Question 33 continues on page 31

Question 33 (continued)

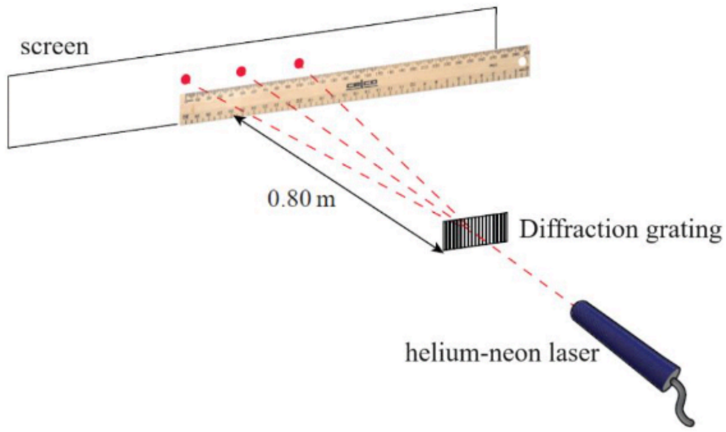
Using the three stimuli provided, show how these three particles improved our understanding of the structure of matter and the laws of physics. **9**

Include a significant calculation for at least two of the three stimuli provided.

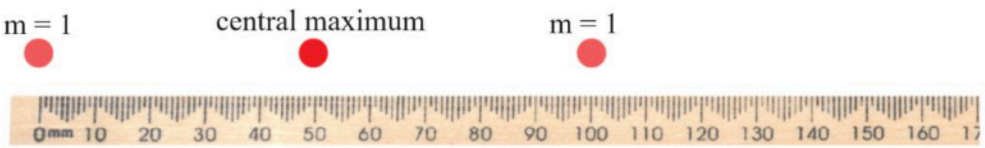
End of Question 33

Question 34 (5 marks)

A student undertook an experiment to investigate diffraction using a helium-neon laser and a diffraction grating containing 100 lines per mm, positioned 0.80 m from a screen.



The image below shows the distances between the central maximum and the first orders of the diffraction pattern that was observed on the screen.



drawn to scale

(a) Calculate the wavelength of the helium-neon laser. **3**

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(b) Given that the range of the visible spectrum is 400 nm to 700 nm, predict how the distances between the central maximum and subsequent maxima will be affected if blue laser light is used in the above investigation. **2**

Justify your answer.

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Question 35 (7 marks)

A particle P, of mass $2m$ and charge $+q$, is injected into an evacuated glass chamber with an initial velocity of v. The particle experiences a uniform magnetic field perpendicular to the direction of motion and a uniform gravitational field directed downwards. Particle P travels through the tube in a straight horizontal path until it reaches X, as shown in Figure 1.

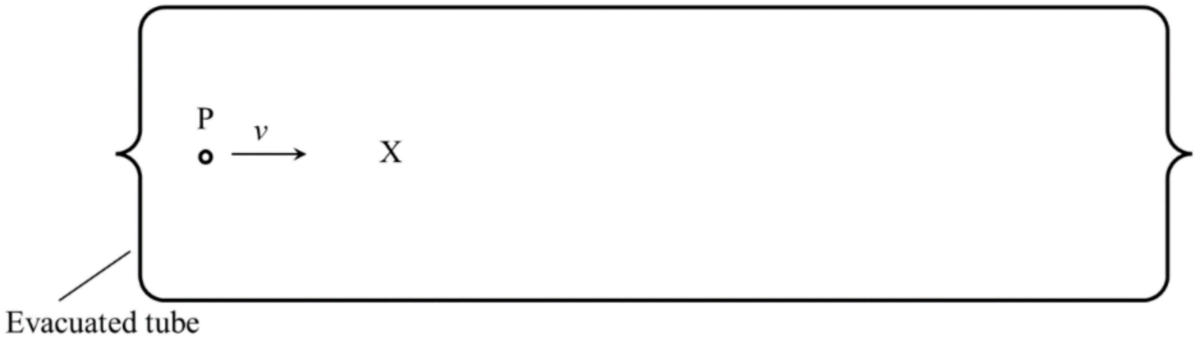

Figure 1

Particle P has a fine crack in it and when it reaches X, it breaks into two fragments Q and R as shown in Figure 2. The upper fragment, Q has mass m, and charge $+q$, and the lower fragment R, also has mass m but is uncharged. The breaking process is elastic and momentum is conserved during the breaking process.

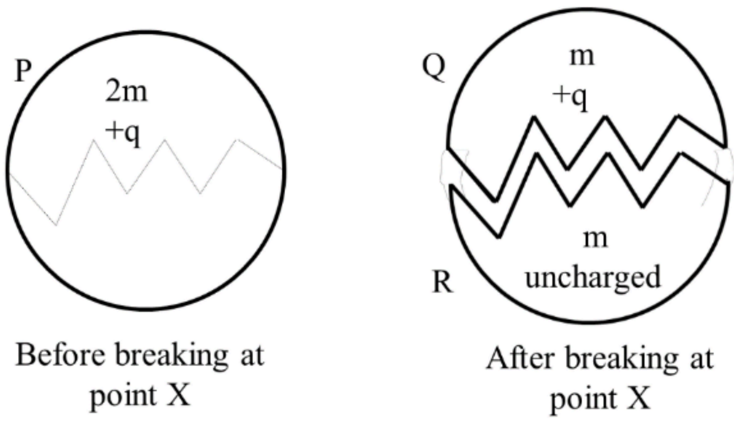

Figure 2

Question 35 continues on page 35

Question 35 (continued)

Account for the trajectories of the initial particle P and its fragment R and discuss factors that affect the trajectory of fragment Q. **7**

Support your answer with mathematical models, relevant laws of physics and diagrams.

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End of Exam

Part B extra writing space

If you use this space, clearly indicate which question you are answering here AND on your paper.